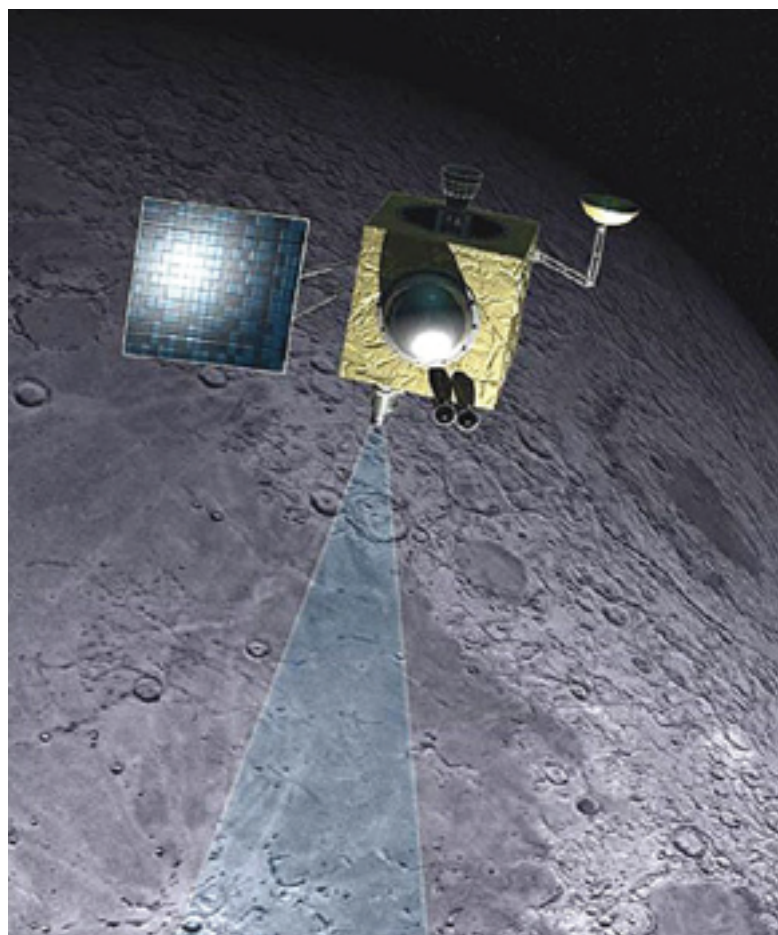


Chandrayaan



Siddharth Parwatay

Chandrayaan-I, India's first mission to the moon was launched on October 22nd, 2008. At the time of going to press, the mission has seen no major glitches and has been near flawless, and we hope it stays that way. It has already beamed back to earth around 40,000 postcard-size high-definition images of the moon. Soon, some 70 scientists from across the globe are expected to pour over these images to unravel the mysteries of the moon.

Overall, the mission had three objectives. The primary objectives of the mission are to expand the scientific knowledge about the origin and evolution of moon, upgrade India's technological capabilities, and provide challenging opportunities to young scientists working in the planetary sciences. These objectives are to be achieved through high resolution remote sensing of the moon

in the visible, near-infrared, microwave and X-ray regions of the electromagnetic spectrum. With this, preparation of a 3D atlas and chemical mapping of entire lunar surface is being carried out. Some really fancy equipment like the Terrain Mapping Camera (TMC), Lunar Laser Ranging Instrument (LLRI), and some other spectral and X-ray imaging instruments were loaded on to the Chandrayaan.

The mission has certainly achieved some remarkable feats. The first one that comes to mind is more economic than scientific. The entire mission was pulled off in a comparatively humble budget of Rs 386 crores. Great! But some questions remain unanswered — why would you want to go to the moon in the first place? Didn't the space race die out with the Cold War? Agreed, the innate human desire for exploration is undeniable, but there have to be some real tangible benefits. Well we decided to find out through a one-on-one round with the scientists at ISRO. And while we were at it, we bombarded them with a host of other questions regarding the technology used, breakthroughs achieved, and future plans.

Digit: What were the motives behind the mission? Why would we need a detailed topographic mapping of the moon, and why analyse the lunar surface for mineral deposits?

ISRO: The primary objectives of Chandrayaan-I are: i) upgrade India's technological capabilities, ii) expand scientific knowledge about the moon, iii) provide challenging opportunities for planetary research to the younger generation of Indian scientists. Outer space will be the next frontier of exploration for mankind. The Moon will form an intermediate base for undertaking exploration of outer space. There is a need to identify the presence of water to support astronauts undertaking deep space flights, and for using water by breaking up into hydrogen and oxygen as a possible source of propellant for future missions. There is also a need to exploit minerals and chemicals available on the lunar surface. Preparation of a detailed topographic map is essential to understand different terrains of the Moon. Chandrayaan-I also aims to look at the presence of Helium 3 on the lunar surface, which can be used as a fuel in nuclear reactors that can be useful in solving the power requirements on Earth.

Digit: What were the kind of mission-specific requirements that were factored in designing the Terrain Mapping Camera? What is the image resolution of on-board cameras like the TMC?

ISRO: The aim of the Terrain Mapping Camera is to completely map the topography of the moon. The camera works in the visible region of the electromagnetic spectrum and captures black-and-white stereo images. It scans a strip of lunar surface which is 20 km wide with a resolution of 5 m. Such high-resolution imaging helps in better understanding of the lunar evolution process, as well as in the detailed study of the regions of scientific interest. When used in conjunction with data from Lunar Laser Ranging Instruments (LLRI), it can help in better understanding of the lunar gravitational fields as well.

Digit: How was the mission accomplished on such a low budget?

ISRO: The total budget for Chandrayaan-1 project was Rs 386 crores, out of which about Rs 100 crores was spent on the Deep Space Network, which has 32-m and 18-m antennas which can be used for Chandrayaan-I as well as future missions. The Chandrayaan-I mission was accomplished within the budget by optimal utilisation of human resources, infrastructure and good planning.

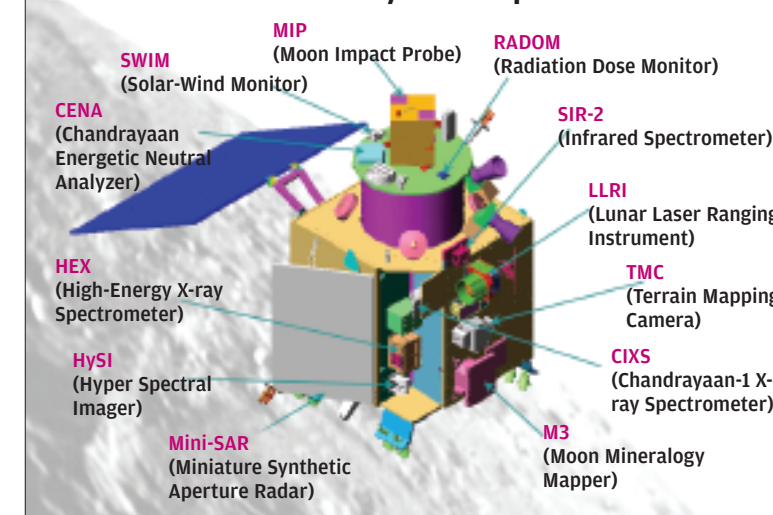
Digit: What exactly is the Indian Deep Space Network (IDSN)? How does it counter any time delays?

ISRO: The Indian Deep Space Network performs the important task of receiving radio signals transmitted by Chandrayaan-I spacecraft that become incredibly feeble by the time they reach the Earth. Besides, it can send commands to the spacecraft at a power level of up to 20 KW. The IDSN consists of two large parabolic antennas, one with an 18-m, and the other 32-m diameter at Byalalu, situated about 35 km from Bangalore. Of these, the 32-m antenna with its 'seven mirror beam waveguide system' was indigenously designed, developed, built, installed, tested and qualified. The 18-m antenna can support the Chandrayaan-I lunar mission, while the 32-m antenna can support any spacecraft mission deeper into space.

Digit: How would the Indian space programme compare to NASA in terms of technology?

ISRO: India is on par with the best in the world in the area of satellite technology. India has the largest constellation of remote sensing and communication satellites in the Asia-Pacific region in civilian sector. ISRO is considered a leader in the application of space technology for the benefit of mankind and national development. The social applications such as tele-medicine and tele-education conceived by ISRO and the setting up of village resource centres,

The Chandrayaan-1 Spacecraft



remote sensing services and communication satellites has drawn the attention the world.

Digit: What kind of future surface missions can we expect post Chandrayaan-I? Is Chandrayaan the first step in setting up a future moon base? Is a mission to Mars on the cards?

ISRO: Following Chandrayaan-I, ISRO has plans to launch Chandrayaan-II around 2011-12, carrying a lander and rover to collect samples on the lunar surface and analyse the chemical composition of the lunar soil. Studies have been initiated for undertaking an unmanned mission to Mars.

Digit: What is the significance of India being one of four countries to have its flag on the moon?

ISRO: When the Moon Impact Probe (MIP) with Indian tricolor painted on its surface impacted the lunar soil, India became the fourth country to land a man-made object on the lunar surface. This is a significant achievement as the mission involved travelling a distance of 400,000 kilometres performing crucial manoeuvres to place the satellite in lunar orbit and impacting lunar soil at predetermined place with great accuracy.

Concluding thoughts

Although we were quite eager to know some details regarding the computer technology used in the mission, such as processor, computing power, AI (if any), unfortunately, ISRO was pretty tight-lipped.

We're assuming that this may be sensitive information, or just national trade secrets, which we obviously do not want to share with the world. Stay tuned to this space, and we will try and get continued updates about Chandrayaan, and future missions, as our nations flag is carried proudly into outer space. ■

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You can't take a trip to the moon just yet, but you can see a few stars on your way to our Mumbai office. Details on Page 78